



Driving ARPU Growth Through Customer Usage Behavior and Plan Optimization: A Telco Analytics Case Study

Customer usage and plan insight TelcoX

Presented by

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Company Overview

TelcoX is a growing telecommunications startup that has been operating for approximately two years, offering mobile data and voice subscription plans to individual customers across multiple regions.

By the end of 2025, the company was facing significant competition from other providers and was keen to implement improvements as a matter of urgency. Meanwhile, the internal team had also conducted an evaluation of the increasingly stagnant Average Revenue Per User (ARPU), and had identified plans which did not meet the standards used by customers.

To address this challenge, TelcoX aims to **leverage customer usage behavior analytics** to improve **plan alignment, enhance customer value**, and **drive sustainable ARPU growth** through data-driven decision making.

Executive Summary

As TelcoX enters its growth phase, the company seeks to improve revenue efficiency by optimizing how customers are aligned with subscription plans based on actual usage behavior.

Key Findings

- Approximately 38% of customers experience data overusage, primarily concentrated among Basic and Standard plan users.
- Customers with frequent overusage generate higher ARPU, but also face increased risk of dissatisfaction due to plan mismatch.
- A significant portion of users exhibit underusage, indicating inefficiencies in current plan allocation.
- Add-on purchases are more common among medium to high usage segments, suggesting strong upsell potential.

Business Implication

- Misaligned plans represent both a revenue growth opportunity and a retention risk if left unaddressed.
- Optimizing plan alignment can improve customer satisfaction while unlocking incremental ARPU growth

	Plan	AVG Usage
Overusage	BASIC	10-12 GB
	STANDARD	15 GB Quota
	UNLIMITED	10 GB
Underusage	PREMIUM	100 GB Quota
	STANDARD	

Recommendation Actions

- Implement personalized plan upgrade recommendations for customers consistently nearing or exceeding their data quota.
- Introduce usage-based nudges to guide underusage customers toward more suitable plans.
- Leverage customer segmentation to design targeted upsell strategies for add-on services.

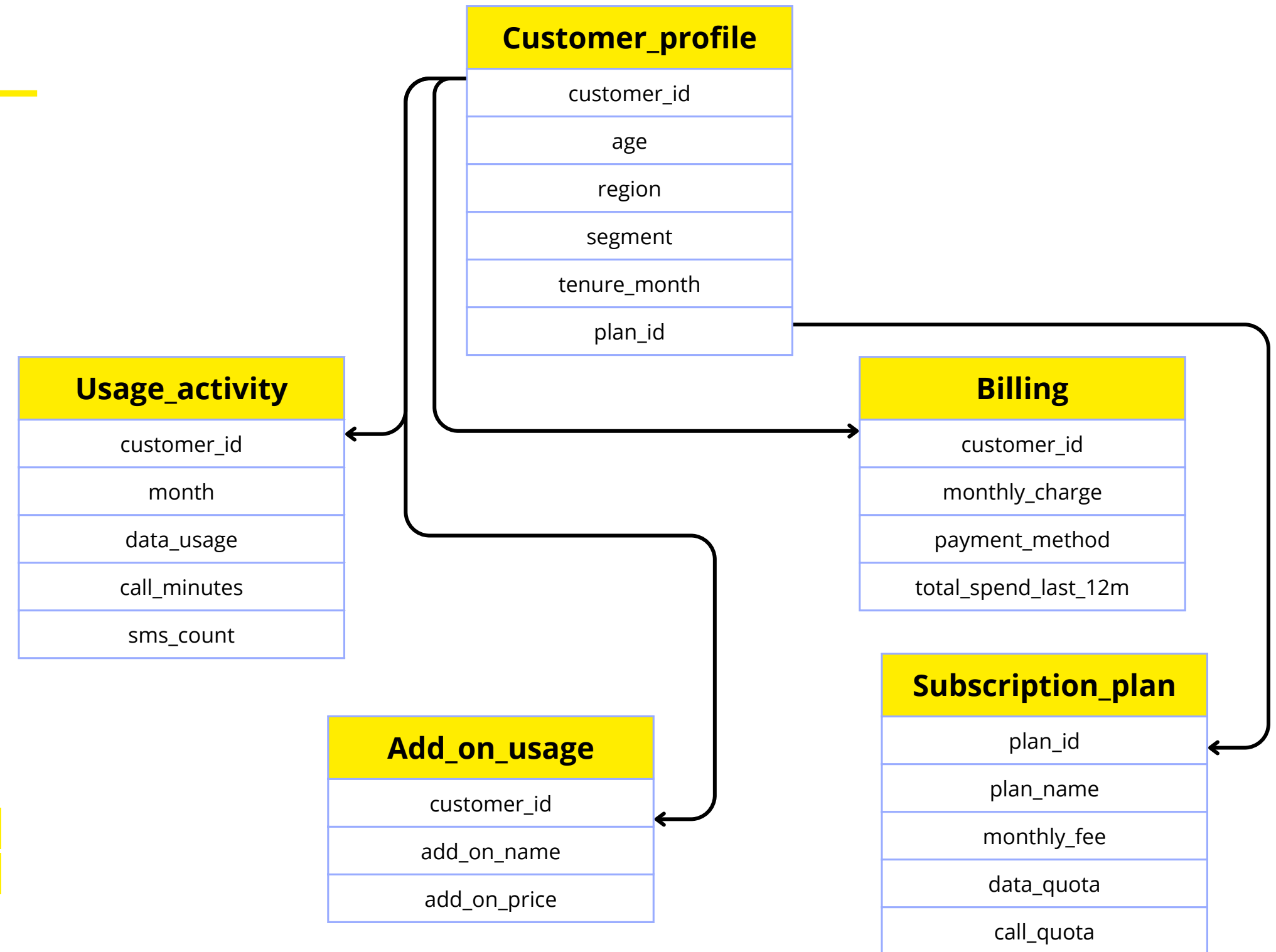
Problem Statement

“How can TelcoX improve ARPU by aligning subscription plan with customer usage behavior?”

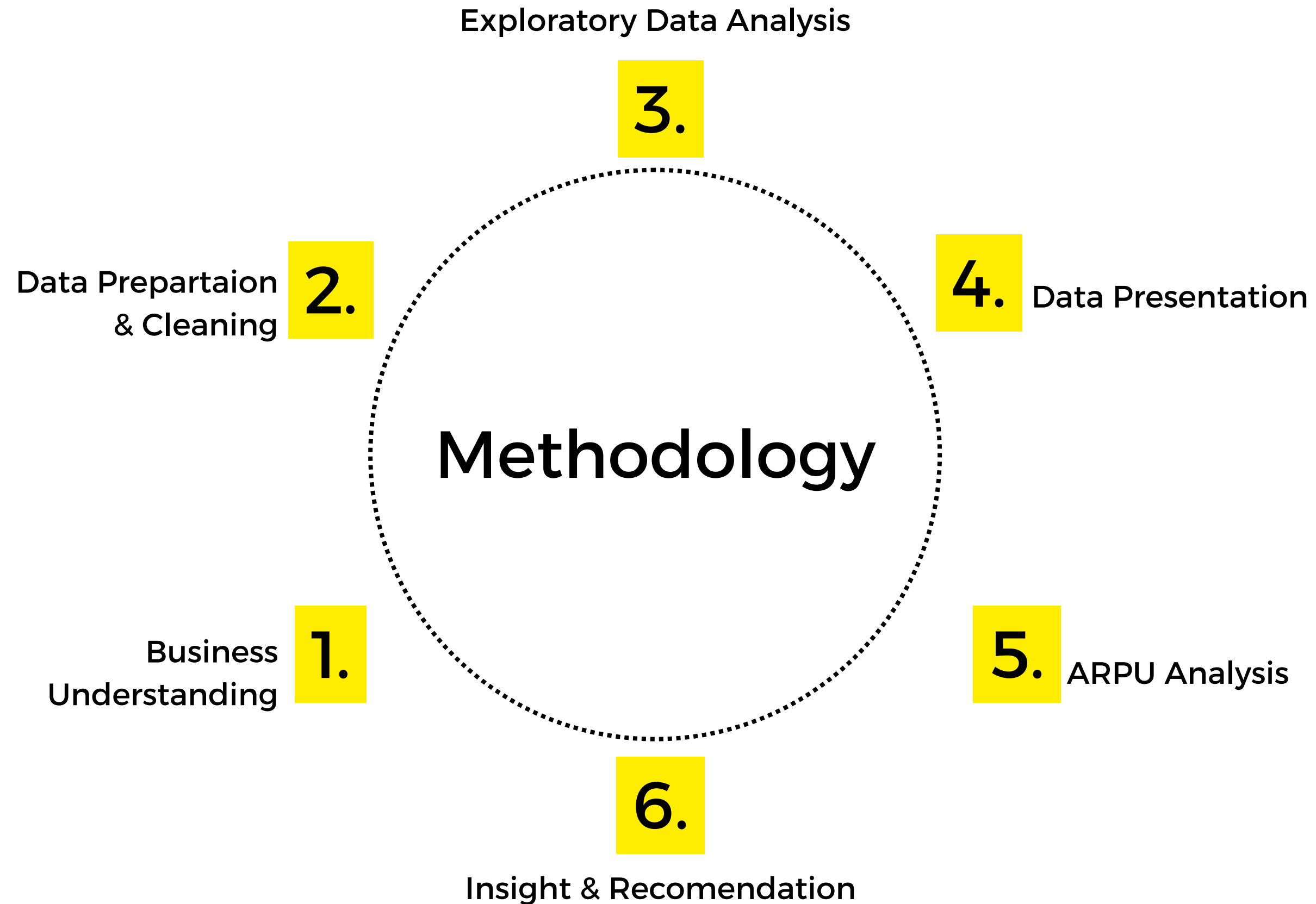
Objective

- Understand customer usage behavior across different subs
- Identify mismatch plan, including overusage and underusage segment
- Evaluate the impact of usage behavior on ARPU and Add-on Purchases
- Provide data-driven recommendations to optimized plan allocation and revenue growth

Data Structure



The TelcoX data team provided a dataset from five different tables, namely **Customer Profile**, **Usage Activity**, **Add on Usage**, **Billing**, and **Subscription Plan**. To assist with the analysis, we will perform data extraction using **SQL Query**.



KPI Overview

Usage Imbalance Signals Revenue Optimization Opportunity

Total Customer

2,000

Average Data Usage

10 GB
/Month

Overusage Rate

38%

Average Revenue Per User
(ARPU)

Rp. 115,753
/Month

Add-on Adoption Rate

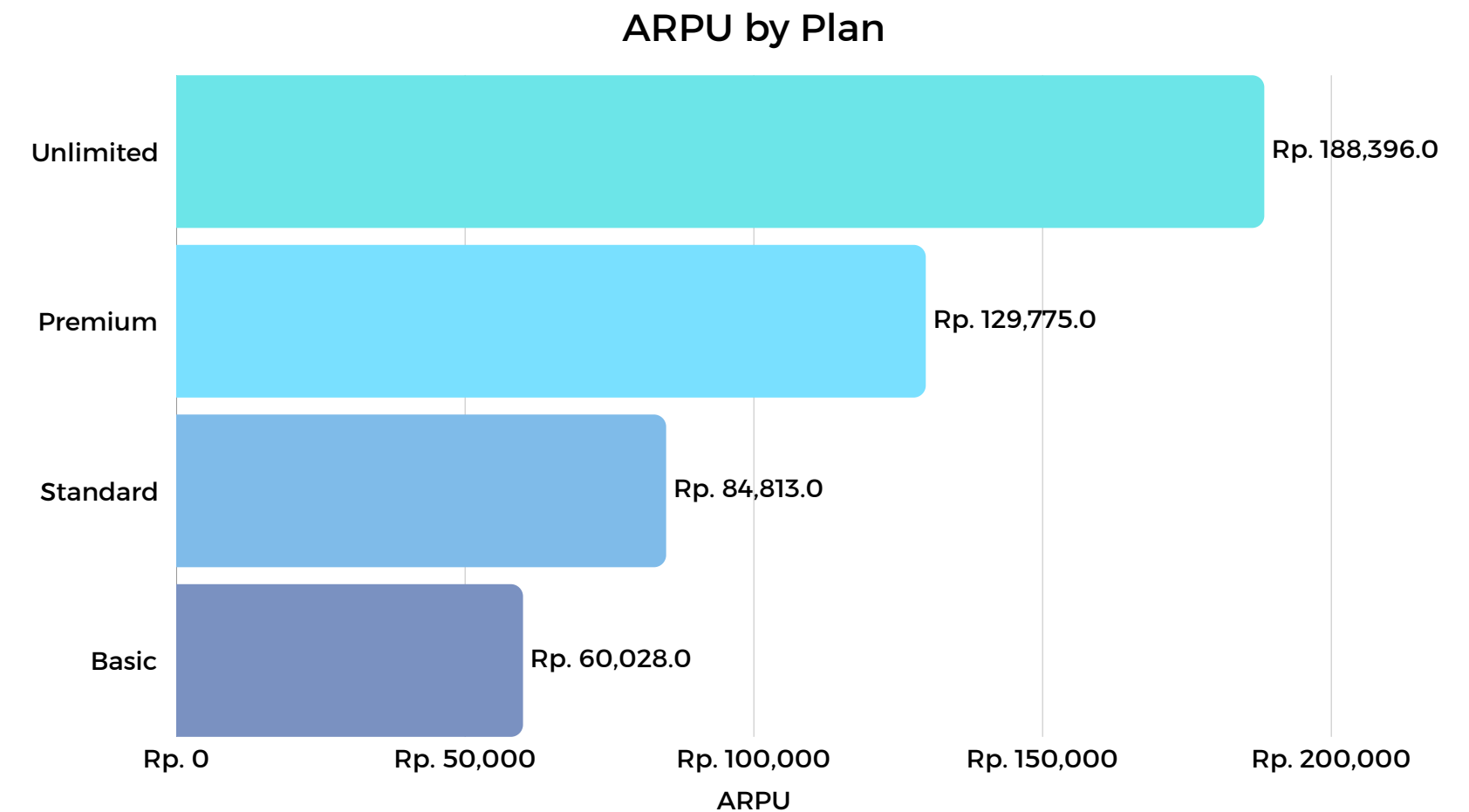
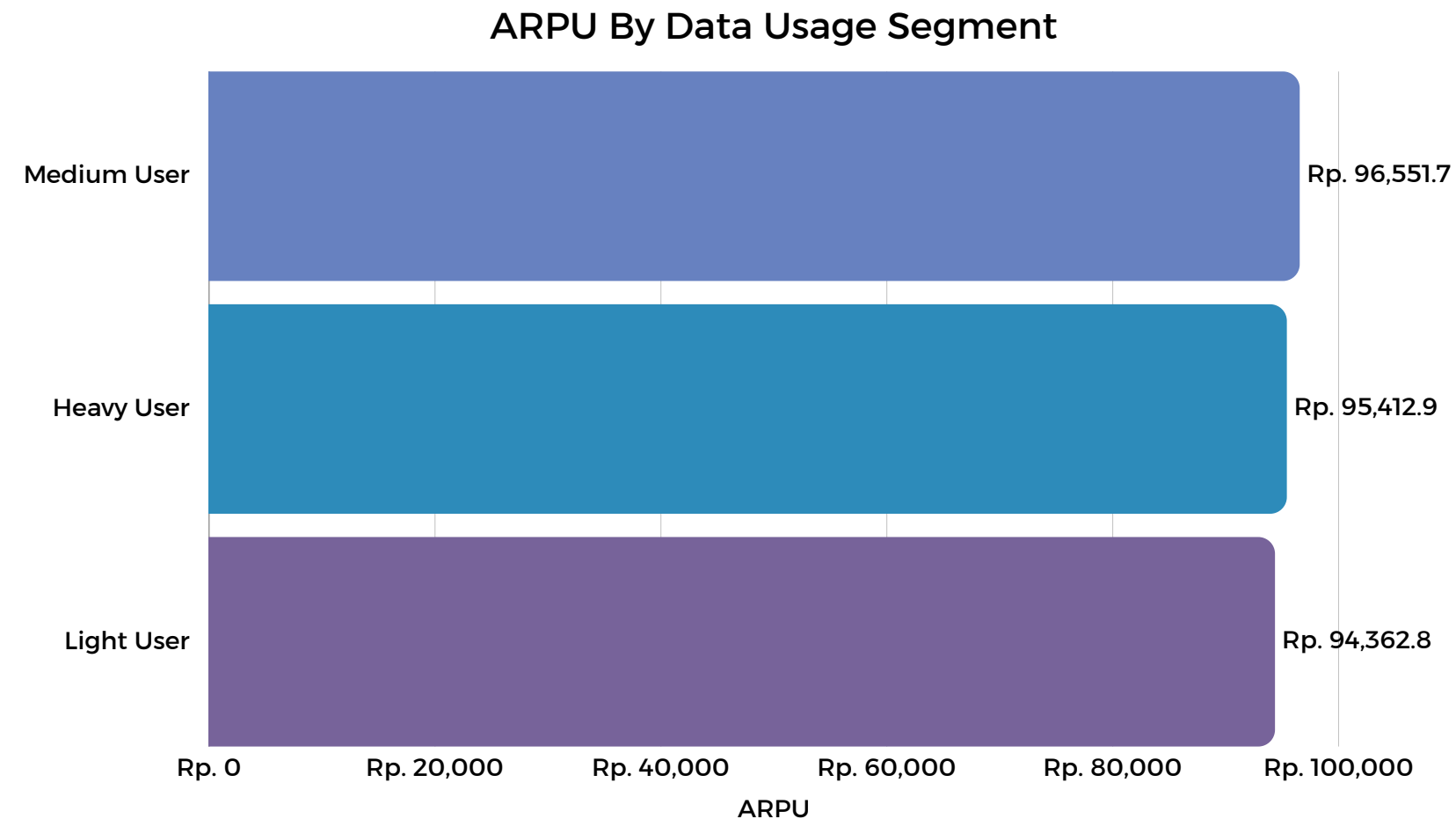
35.6%

Key Findings

- A high overusage rate (38%) suggests a mismatch between customer needs and current subscription plans
 - Despite stable ARPU, usage patterns indicate untapped upsell potential, particularly among heavy users.
 - Add-on adoption remains moderate, highlighting opportunities for targeted cross-selling strategies.
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ARPU Analysis

Higher Usage Customers Drive ARPU Disproportionately Across Plans

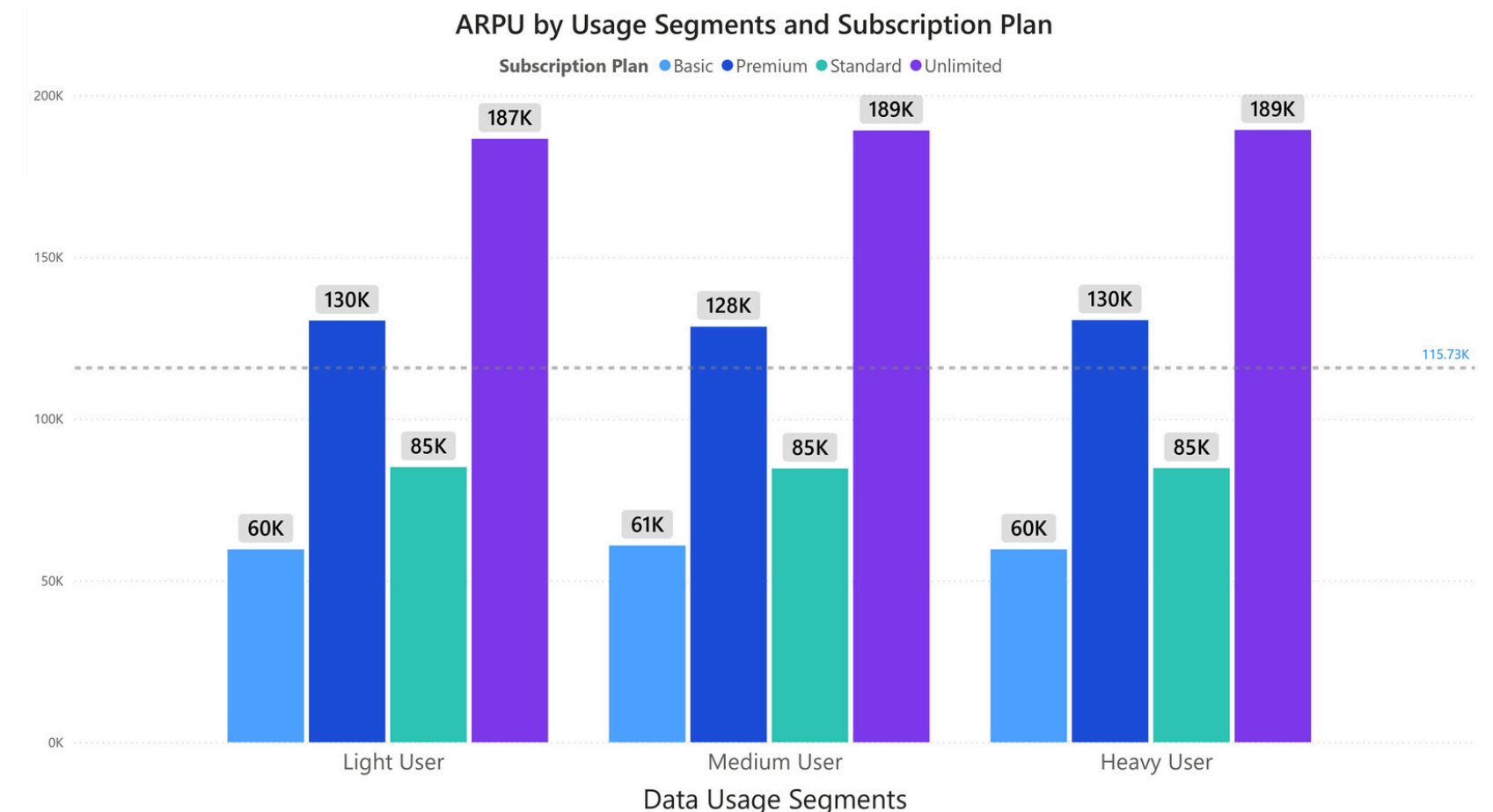
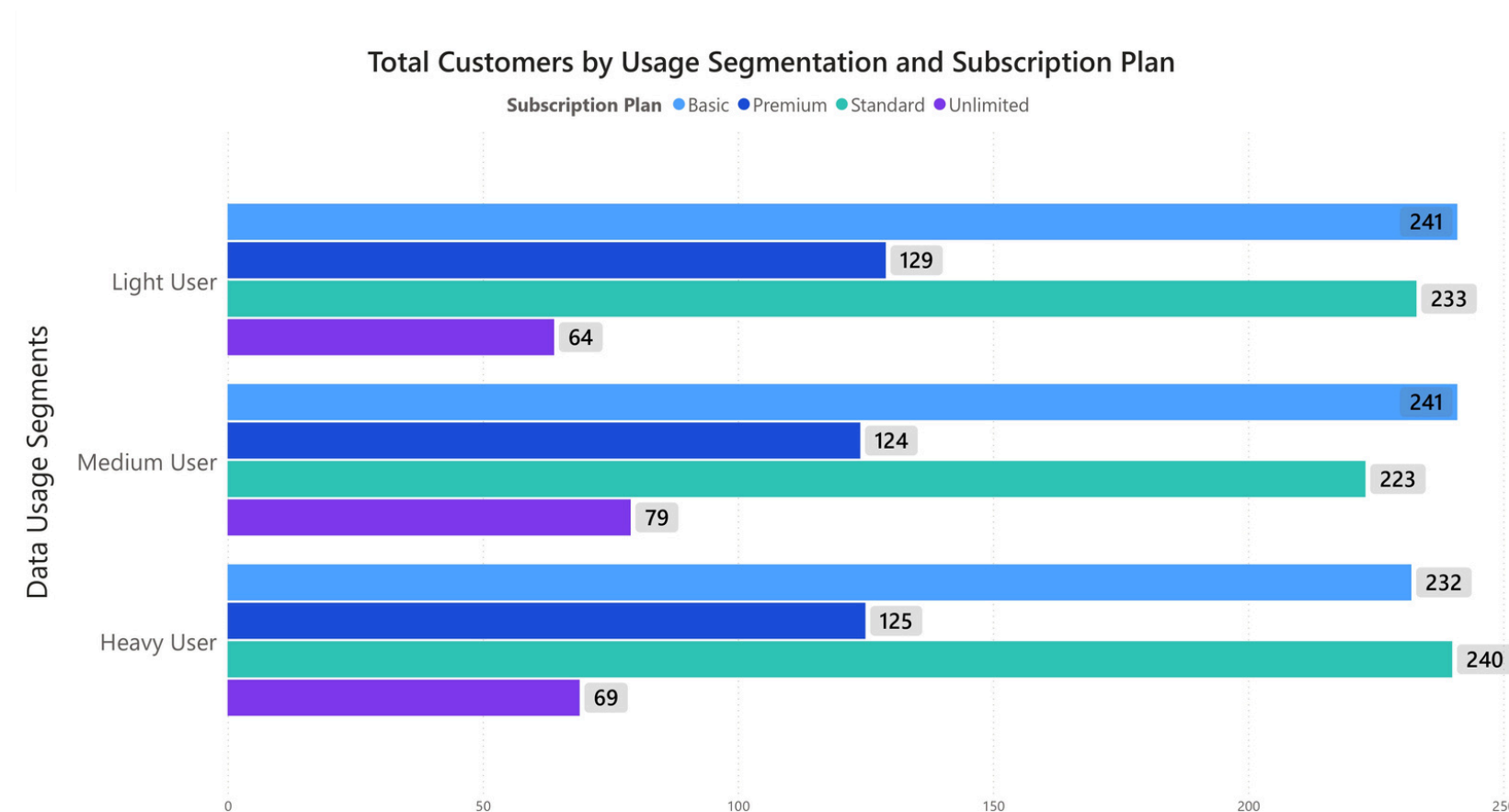


Key Findings

- Despite large differences in usage intensity, ARPU remains relatively flat across segments.
- Heavy users represent latent monetization opportunity rather than current revenue drivers.
- Revenue is more likely driving by plan distribution and not by data usage.

ARPU Analysis – Revenue Distribution by Usage & Plan

Potential upselling between usage behavior with looking what plan that customer use.

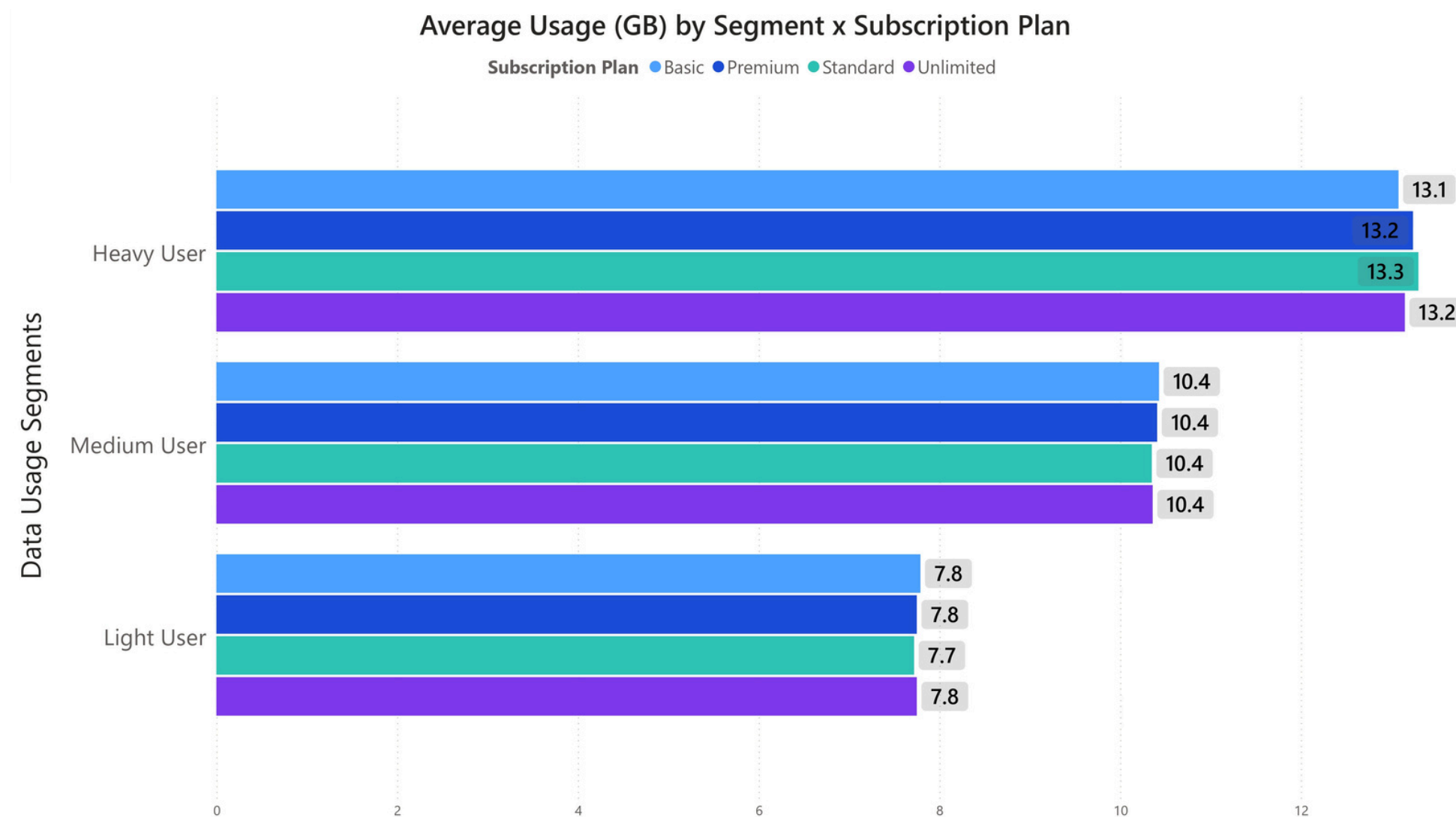


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Usage vs Plan Alignment

Mismatch Between Usage Behavior and Plan Alignment



Each plan have a different limited quota. For **Unlimited** you have 100 GB, **Premium** you have 30 GB, **Standard** have 15 GB, and **Basic** have 5 GB. This analysis reveals a consistent **mismatch** between customer data usage and subscribed plan quotas across all usage segments. This misalignment results in unrealized ARPU potential for high-usage customers and inefficiencies in plan allocation for low-usage customers.

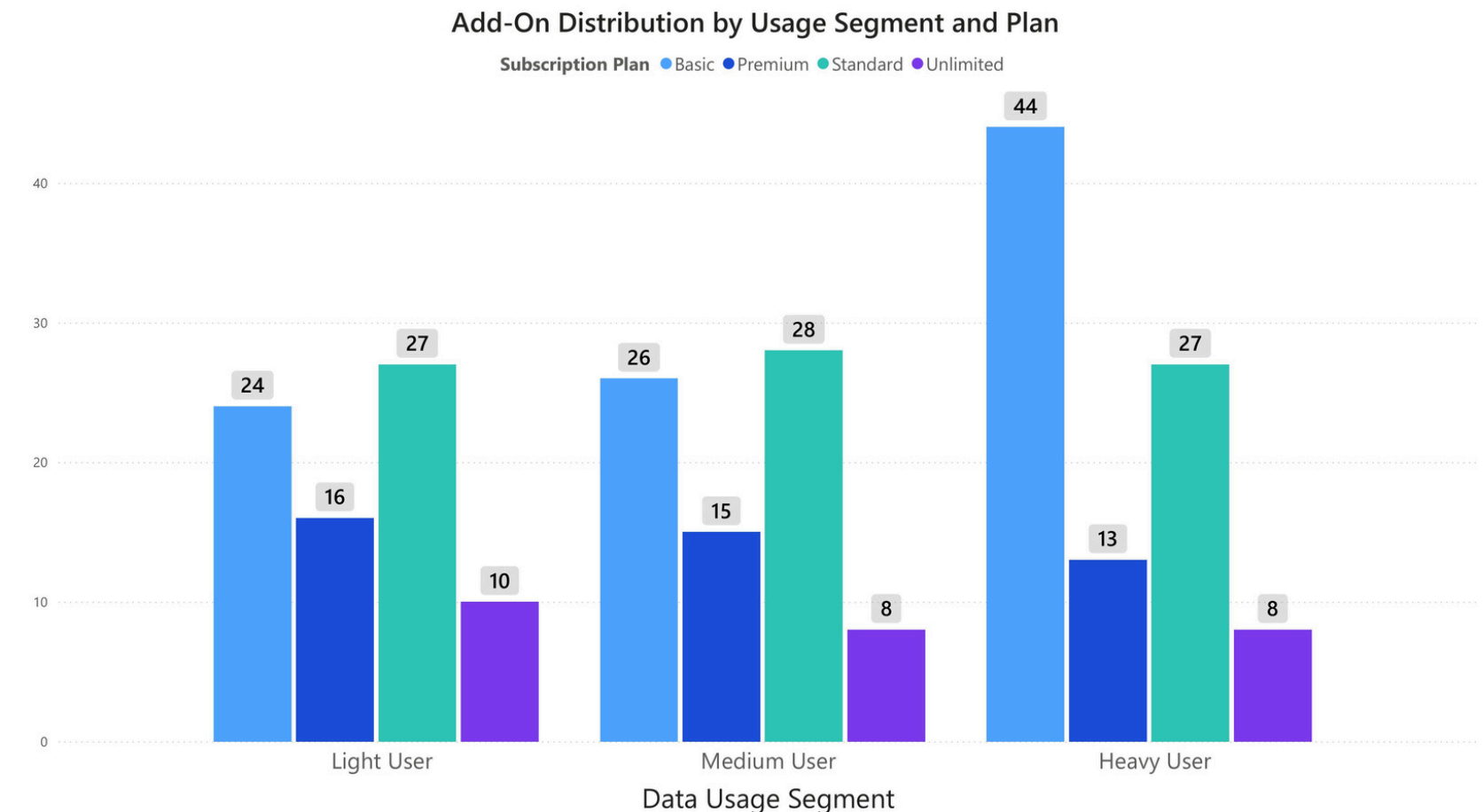
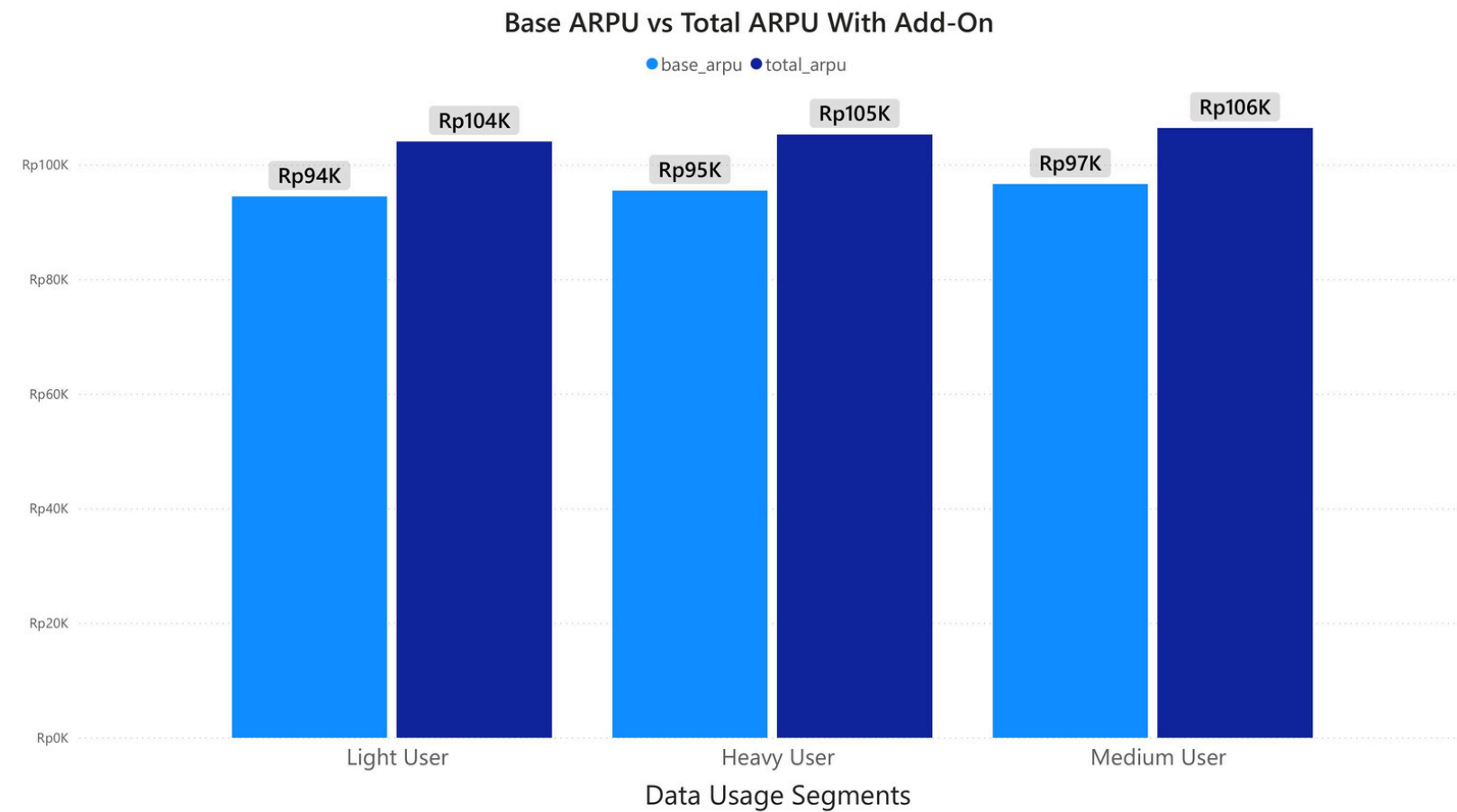
Key Findings

- Heavy users under-subscribed relative to usage
- Medium users clustered near Standard threshold
- Light users over-allocated in higher quota plans

This misalignment indicates both upsell and cost optimization opportunities.

ARPU Analysis – Add-On Revenue by Usage and Plan

It is important to acknowledge that customers have different needs, which often require them to purchase additional.

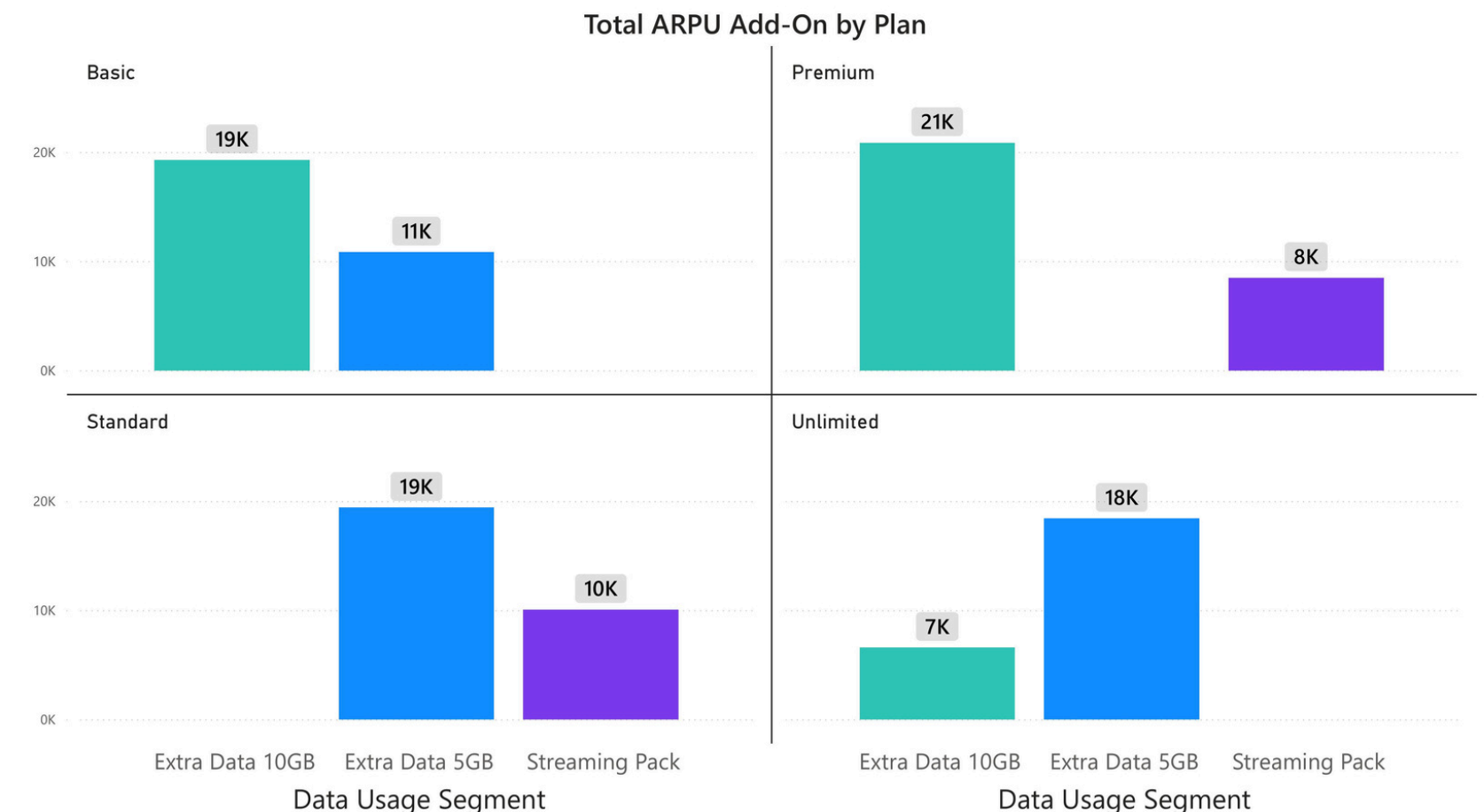
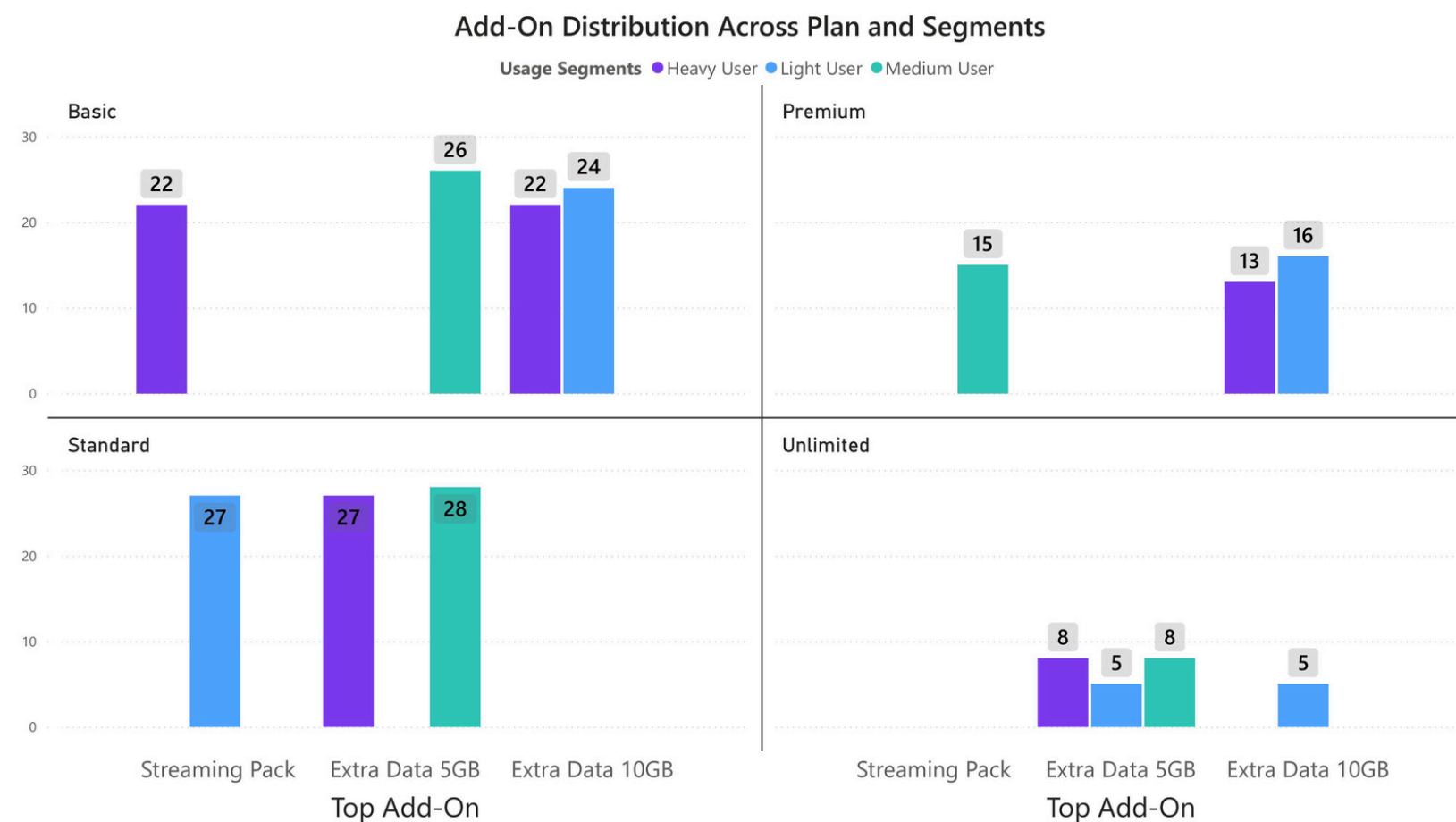


Key Findings

- The introduction of the add-on has led to an increase in ARPU from the ARPU base.
- This indicates that certain customer needs may not be fully addressed by their current plan.
- It is evident that a significant proportion of customers who do not align with the plan are likely to purchase the add-on.
- Customers subscribed to the Basic and Standard plans purchase add-ons at a high frequency.

ARPU Analysis – Add-On Revenue by Usage and Plan

A quick look distribution of total customer, avg age, segment, and plan.



Key Findings

- Add-on sell is helping all of the customer on plan to fulfill they need.
- Heavy user are the biggest buyer on basic plan indicating that most of them is in the wrong plan.
- All customer in every plan are needed the Extra GB.
- The Streaming Pack can be the solution to people who have the streaming behavior. especially on the Standard and Premium plans.

Key Insights

Key Insights from Usage & Revenue Analysis

- A significant portion of high-usage customers remain on lower-tier plans, indicating unrealized ARPU potential.
- Medium usage customers show strong add-on adoption, suggesting readiness for plan upgrades or bundled offerings.
- Light usage customers are frequently over-allocated in higher quota plans, creating cost inefficiency without revenue upside.
- Add-on purchases act as a revenue bridge, signaling unmet needs in existing subscription plans.

Business Recommendations

Targeted Revenue & Retention Strategies

- 1. Upsell High-Usages Customer**
Prioritize proactive upgrades to Premium or Unlimited plans for customers consistently exceeding quota.
- 2. Convert Medium Users via Bundled Offers**
Introduce data + add-on bundles to transition medium users toward higher-value plans.
- 3. Optimize Plans for Light Users**
Encourage right-sizing through standard plans to improve perceived value and reduce network inefficiency.
- 4. Leverage Add-On Behavior as Upsell Signals**
Use frequent add-on purchases as indicators for plan mismatch and upgrade readiness.
- 5. Deploy Usage-Based Nudges**
Implement data usage notifications to guide customers toward optimal plans before overusage occurs.

Potential Impact

Estimated Business Impact (Simulation)

- If 20% of misaligned heavy users upgrade to higher-tier plans, TelcoX could unlock incremental ARPU growth.
- Upselling medium users through bundled plans may increase add-on conversion while reducing overusage friction.
- Plan optimization for light users can improve customer satisfaction and reduce unnecessary network cost.

"Impact estimates are based on observed usage distribution and ARPU differentials."

Disclaimer (dummy data)

This project uses synthetic and anonymized data created for portfolio demonstration purposes. The analysis methodology and insights are designed to reflect real-world telco analytics use cases. Findings should be interpreted as directional insights rather than actual business performance metrics.

Worksheet

SQL Worksheet